

学生问题:

1. $\begin{pmatrix} \\ \\ \end{pmatrix} \rightarrow \begin{pmatrix} \\ \\ \end{pmatrix}$

2. 求解线性方程组的几何解释

矩阵初等变换:

[Systems of Linear Equations \(gatech.edu\)](https://www.gatech.edu)

三、用矩阵的初等行变换解方程组

$$B = \begin{pmatrix} 2 & -1 & -1 & 1 & 2 \\ 1 & 1 & -2 & 1 & -4 \\ 4 & -6 & 2 & -2 & 4 \\ 3 & 6 & -9 & 7 & 9 \end{pmatrix}$$

$$\begin{array}{l} r_1 \leftrightarrow r_2 \\ r_3 \div 2 \end{array} \begin{pmatrix} 1 & 1 & -2 & 1 & 4 \\ 2 & -1 & -1 & 1 & 2 \\ 2 & -3 & 1 & -1 & 2 \\ 3 & 6 & -9 & 7 & 9 \end{pmatrix} = B_1$$

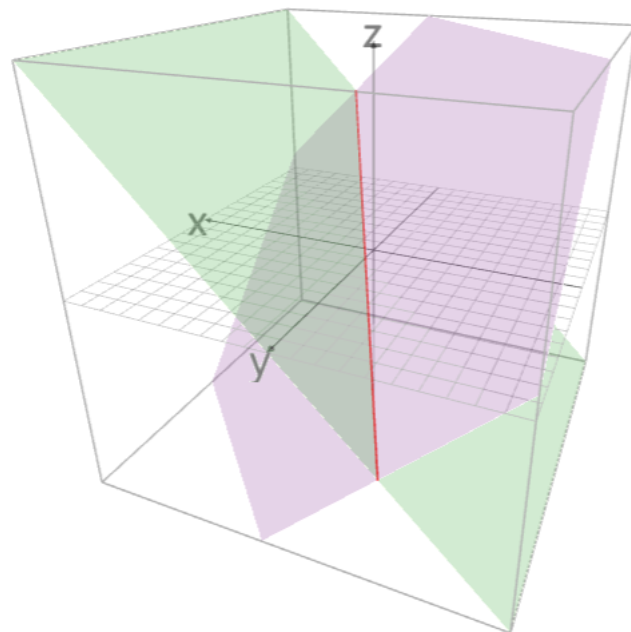
$$B_1 = \begin{pmatrix} 1 & 1 & -2 & 1 & 4 \\ 2 & -1 & -1 & 1 & 2 \\ 2 & -3 & 1 & -1 & 2 \\ 3 & 6 & -9 & 7 & 9 \end{pmatrix} \begin{array}{l} r_2 - 2r_1 \\ r_3 - 2r_1 \\ r_4 - 3r_1 \end{array} \begin{pmatrix} 1 & 1 & -2 & 1 & 4 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & -5 & 5 & -3 & -6 \\ 0 & 3 & -3 & 4 & -3 \end{pmatrix} = B_2$$

$$\begin{array}{l} r_2 \div 2 \\ r_3 + 5r_2 \\ r_4 - 3r_2 \end{array} \begin{pmatrix} 1 & 1 & -2 & 1 & 4 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -6 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix} = B_3$$

$$\begin{cases} x + y + z = 1 \\ x - z = 0 \end{cases}$$

Each equation individually defines a plane in space. The solutions of the system of both equations are the points that lie on both planes. We can see in the picture below that the planes intersect in a line. In particular, this system has infinitely many solutions.

$$\begin{cases} x + y + z = 1 \\ x - z = 0 \end{cases}$$



The planes defined by the equations $x + y + z = 1$ and $x - z = 0$ intersect in the red line, which is the solution set of the system of both equations.



parametric form with one *parameter* t :

$$(x, y, z) = (t, 1 - 2t, t).$$

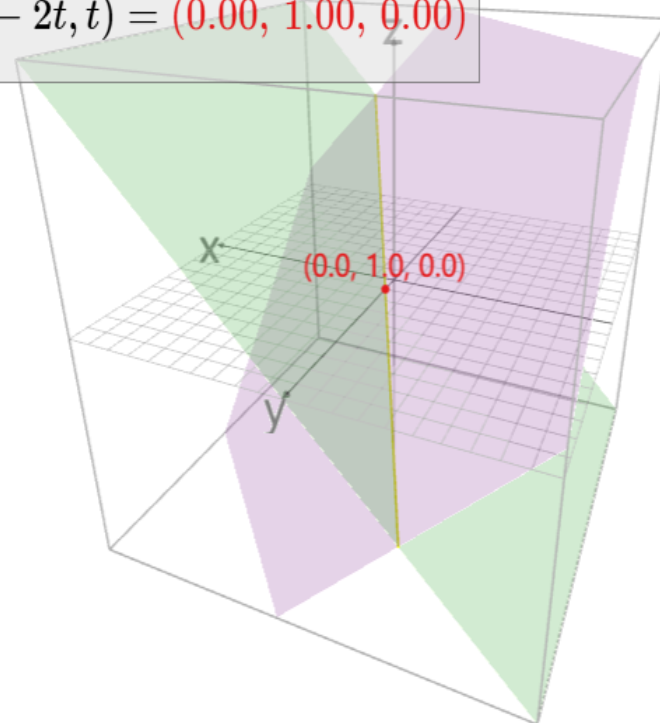
$$\begin{cases} x + y + z = 1 \\ x - z = 0 \end{cases}$$

$$(x, y, z) = (t, 1 - 2t, t) = (0.00, 1.00, 0.00)$$

z

0

Close Controls



The planes defined by the equations $x + y + z = 1$ and $x - z = 0$ intersect in the yellow line, which is parameterized by $(x, y, z) = (t, 1 - 2t, t)$. Move the slider to change the parameterized point.

矩阵 B_4 和 B_5 都称为行阶梯形矩阵.

特点:

(1) 零行在下方

(2) 主元前面的零的
个数逐行严格增加

主元

左起的第一个非零元

$$\begin{pmatrix} \textcircled{1} & 0 & -1 & 0 & 4 \\ 0 & \textcircled{1} & -1 & 0 & 3 \\ 0 & 0 & 0 & \textcircled{1} & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = B_5$$

$$\begin{pmatrix} \textcircled{1} & 1 & -2 & 4 & 4 \\ 0 & \textcircled{1} & -1 & 1 & 0 \\ 0 & 0 & 0 & \textcircled{1} & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = B_4$$

主变量

自由未知量

行阶梯形矩阵 B_5 还称为行最简形矩阵,

(1) 主元为 1;

(2) 主元所在的列的其它元素都为零.

$$\begin{pmatrix} \textcircled{1} & 0 & -1 & 0 & 4 \\ 0 & \textcircled{1} & -1 & 0 & 3 \\ 0 & 0 & 0 & \textcircled{1} & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = B_5$$

Gauss-Jordan算法

增广矩阵



初等行变换

行最简形矩阵



求解



Gauss-Jordan算法

增广矩阵



初等行变换

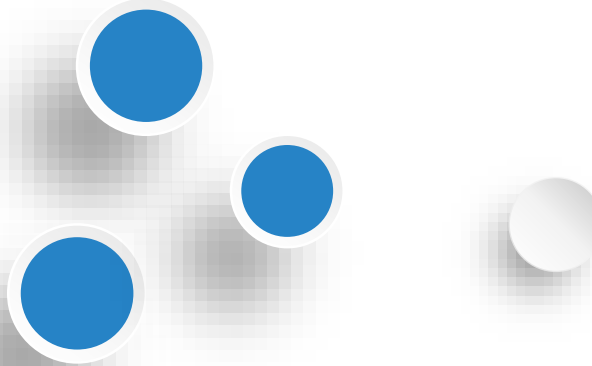
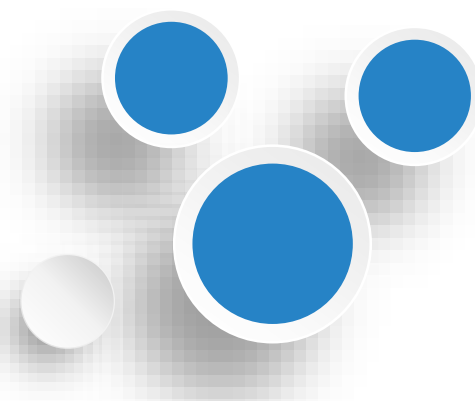
行最简形矩阵



求解

对于任何矩阵,总可经过有限次初等行变换将其变为行阶梯形和行最简形.

试试证明?



§2

线性方程组解的判别



$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

讨论:

$$\bar{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \rightarrow \cdots \rightarrow \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1r} & \cdots & b_{1n} & d_1 \\ 0 & b_{22} & \cdots & b_{2r} & \cdots & b_{2n} & d_2 \\ \vdots & \vdots & & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & b_{rr} & \cdots & b_{rn} & d_r \\ 0 & 0 & \cdots & 0 & \cdots & 0 & d_{r+1} \\ \vdots & \vdots & & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$



结论:

Case1. $d_{r+1} \neq 0$, 无解;

Case2. $d_{r+1} = 0$, 主元个数为 r ($\leq n$)

(1) $r = n$,

讨论:

$$\bar{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \rightarrow \cdots \rightarrow \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1r} & \cdots & b_{1n} & d_1 \\ 0 & b_{22} & \cdots & b_{2r} & \cdots & b_{2n} & d_2 \\ \vdots & \vdots & & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & b_{nn} & d_n \\ 0 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$



$$\begin{pmatrix} 1 & 0 & \cdots & 0 & 0 & d_1 \\ 0 & 1 & \cdots & 0 & 0 & d_2 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & d_{n-1} \\ 0 & 0 & \cdots & 0 & 1 & d_n \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{pmatrix}$$



$$\begin{cases} x_1 = d_1 \\ x_2 = d_2 \\ \cdots \\ x_n = d_n \end{cases}$$



结论:

Case1. $d_{r+1} \neq 0$, 无解;

Case2. $d_{r+1} = 0$, 主元个数为 r ($\leq n$)

(1) $r = n$, 唯一解;

(2) $r < n$,



$$\begin{pmatrix} 1 & 0 & \cdots & 0 & c_{1r+1} & \cdots & c_{1n} & k_1 \\ 0 & 1 & \cdots & & c_{2r+1} & \cdots & c_{2n} & k_2 \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & c_{rr+1} & \cdots & c_{rn} & k_r \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$



$$\begin{cases} x_1 = k_1 - c_{1r+1}x_{r+1} - c_{1r+2}x_{r+2} - \cdots - c_{1n}x_n \\ x_2 = k_2 - c_{2r+1}x_{r+1} - c_{2r+2}x_{r+2} - \cdots - c_{2n}x_n \\ \dots\dots\dots \\ x_r = k_r - c_{rr+1}x_{r+1} - c_{rr+2}x_{r+2} - \cdots - c_{rn}x_n \end{cases}$$



结论:

Case1. $d_{r+1} \neq 0$, 无解;

Case2. $d_{r+1} = 0$, 主元个数为 r ($\leq n$)

(1) $r = n$, 唯一解;

(2) $r < n$, 无穷多解;

定理1

一个 n 元线性方程组的解的情况有且只有三种情况：

无解，有唯一解，无穷多解.

把线性方程组的增广矩阵经过初等行变换化成的行阶梯形矩阵，

- 1) 如果系数矩阵和增广矩阵对应的行阶梯形矩阵的非零行个数不相等，则方程组无解；
- 2) 如果非零行的个数均为 r ,
 - (1) $r = n$, 有唯一解;
 - (2) $r < n$, 无穷多解.

齐次线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0 \\ \dots\dots\dots \\ a_{s1}x_1 + a_{s2}x_2 + \dots + a_{sn}x_n = 0 \end{cases}$$

非零解?



推论1 n 元齐次线性方程组有非零解

$\iff$ 系数矩阵的行阶梯形矩阵的非零行个数 $r < n$.

推论2 n 元齐次线性方程组方程的个数 $< n$

$\implies$ 方程组有非零解.

增广矩阵内在性质?

$$\begin{cases} \alpha_{11}x_1 + \alpha_{12}x_2 + \dots + \alpha_{1n}x_n = b_1 \\ \alpha_{21}x_1 + \alpha_{22}x_2 + \dots + \alpha_{2n}x_n = b_2 \\ \dots\dots\dots \\ \alpha_{s1}x_1 + \alpha_{s2}x_2 + \dots + \alpha_{sn}x_n = b_s \end{cases}$$

线性运算

01

存在性：有没有解，有解的条件？

02

计数：若有解，有多少个？

03

解的结构：若解不唯一，解之间有何关系？如何表示？

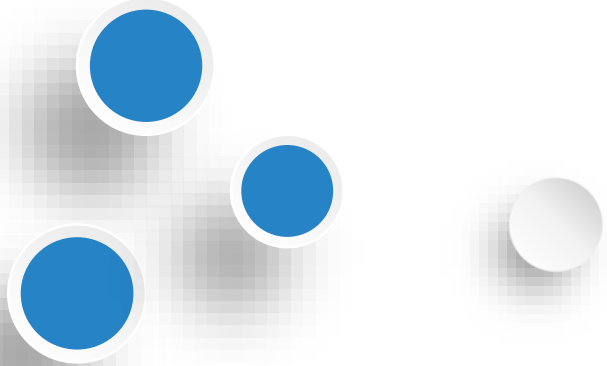
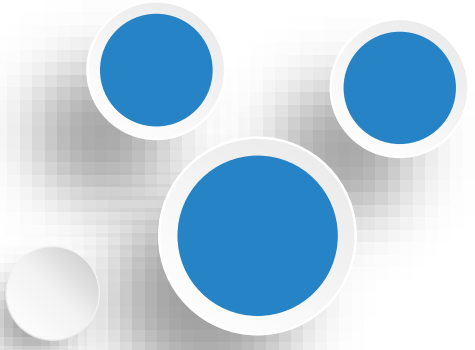

$$\begin{cases} x_1 + x_2 + x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 2 \\ 2x_1 + 3x_2 + 4x_3 = 3 \end{cases}$$



作业：习题1.1: 3(1), (2)

习题1.2: 1, 2, 4

注：只需判解



§3

数域

- 
1. 封闭: 如果数的集合 P 中的任意两个数做某一运算, 结果还在集合 P 中, 则称 P 对这个运算是封闭的.
 2. 数域: 称数的集合 P 为一个数域, 如果 $0, 1 \in P$, 且 P 对加法, 减法, 乘法和除法 (除数不为0) 封闭.

例1. 判断：复数集C 实数集R 有理数集Q 整数集Z

例2. 求证： $Q(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \text{ 为有理数}\}$ 是数域

例3. 设F是一个数集，F至少包含一个不等于0的数，
并且满足：对于 $\forall a, b \in F$, 有 $a - b \in F$,
且对 $b \neq 0$, $\frac{a}{b} \in F$. 证明: F 是一个数域.



定理1: 任何数域都包含有理数域.

◆ 有理数域是最小的数域.

◆ 复数域是最大的数域.

线性方程组

01

行列式

02

向量空间

03



04

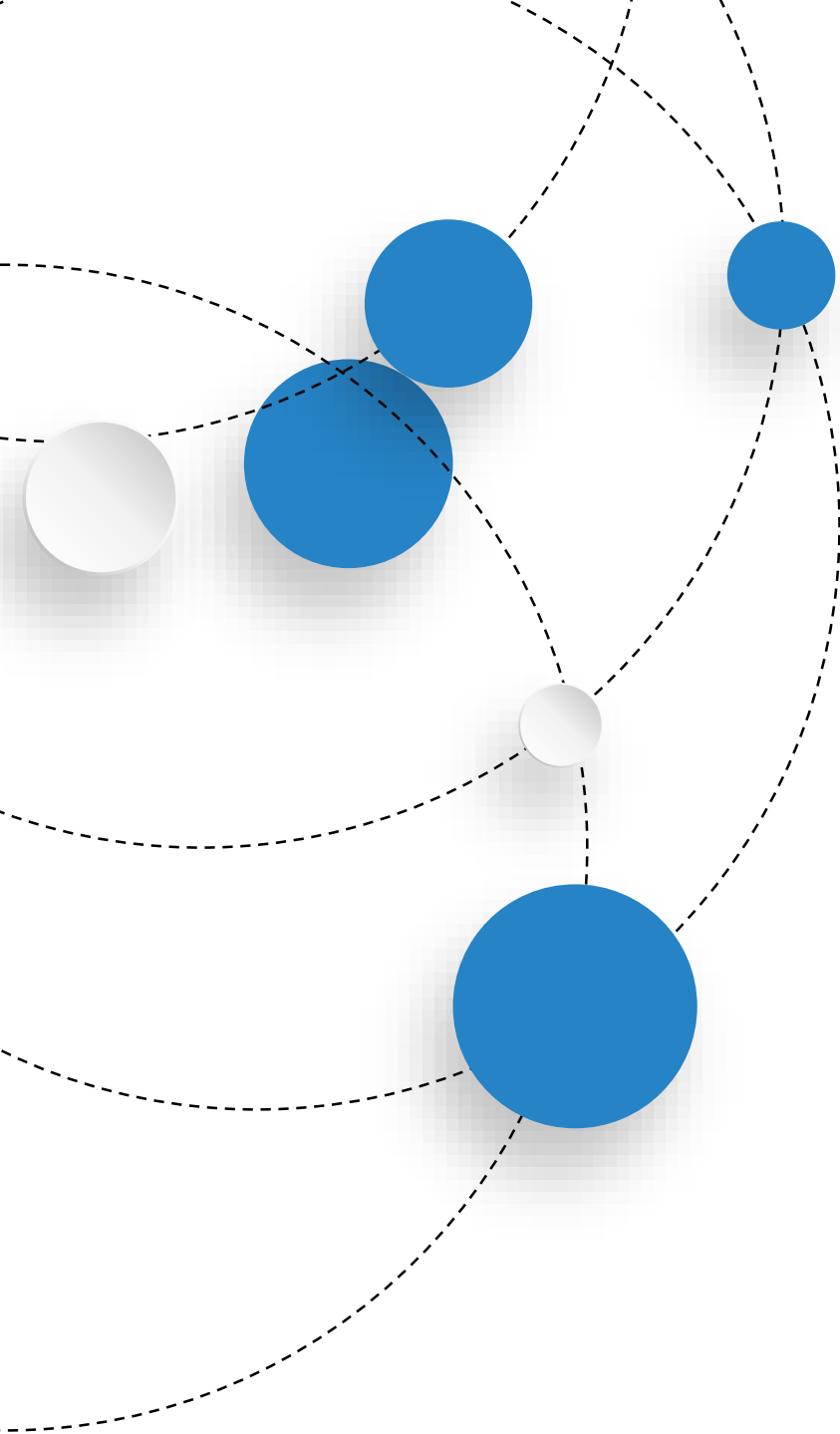
矩阵的运算

05

矩阵的相抵与相似

06

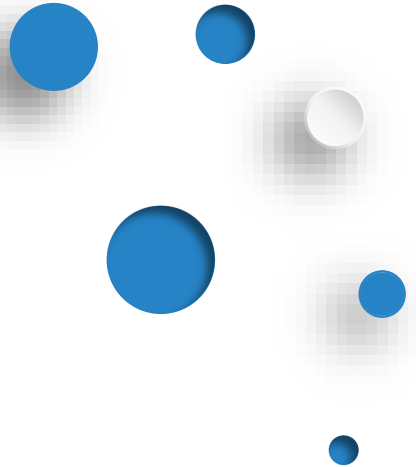
二次型



Ch2

行列式

Determinant



行列式



- 1683年关孝和（日本），1693年Leibnitz(莱布尼兹，德国)



解线性方程组

- 1750年 Cramer（克莱姆，瑞士）

行列式：有向面积或体积的概念在一般的欧几里得空间中的推广
或者说，在 n 维欧几里得空间中，行列式描述的是一个线性变换对
“体积”所造成的影响

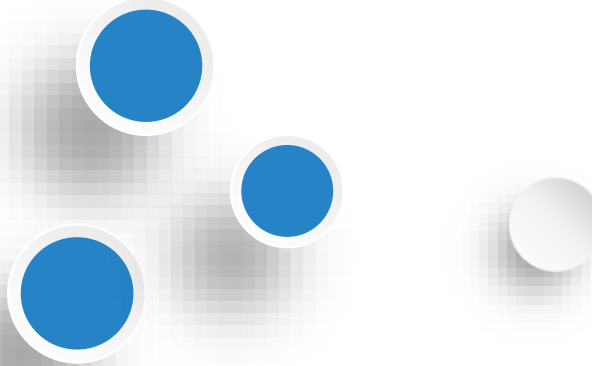
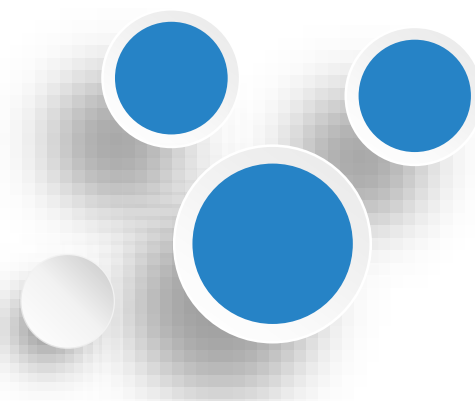
<https://www.bilibili.com/video/BV1V7411z7Jx?from=search&seid=4439901113373228492>

Wronskian (朗斯基) 行列式: 判别函数的相关性——微分方程

Jacobian行列式: n 个 n 元函数的偏导数为元素的行列式——数学分析

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- 排列
- n 阶行列式定义
- 行列式的性质
- 行列式的计算
- 行列式展开
- Cramer法则



§ 1

二阶与三阶行列式

一、二元线性方程与二阶行列式

用消元法解二元线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, & (1) \\ a_{21}x_1 + a_{22}x_2 = b_2. & (2) \end{cases}$$

$$(1) \times a_{22} : \quad a_{11}a_{22}x_1 + \boxed{a_{12}a_{22}}x_2 = b_1a_{22},$$

$$(2) \times a_{12} : \quad a_{12}a_{21}x_1 + \boxed{a_{12}a_{22}}x_2 = b_2a_{12},$$

两式相减消去 x_2 , 得

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, & (1) \\ a_{21}x_1 + a_{22}x_2 = b_2. & (2) \end{cases}$$

$$(a_{11}a_{22} - a_{12}a_{21})x_1 = b_1a_{22} - a_{12}b_2;$$

类似地，消去 x_1 ，得

$$(a_{11}a_{22} - a_{12}a_{21})x_2 = a_{11}b_2 - b_1a_{21},$$

当 $a_{11}a_{22} - a_{12}a_{21} \neq 0$ 时，方程组的解为

$$x_1 = \frac{b_1a_{22} - a_{12}b_2}{a_{11}a_{22} - a_{12}a_{21}}, \quad x_2 = \frac{a_{11}b_2 - b_1a_{21}}{a_{11}a_{22} - a_{12}a_{21}}. \quad (3)$$



引入记号:

二阶行列式

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

元素

行标

列标

$$= a_{11}a_{22} - a_{12}a_{21}$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, & (1) \\ a_{21}x_1 + a_{22}x_2 = b_2. & (2) \end{cases}$$

$$x_1 = \frac{b_1a_{22} - a_{12}b_2}{a_{11}a_{22} - a_{12}a_{21}},$$

$$x_2 = \frac{a_{11}b_2 - b_1a_{21}}{a_{11}a_{22} - a_{12}a_{21}}.$$

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}},$$

$$x_2 = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}.$$

求解二元线性方程组
$$\begin{cases} 3x_1 - 2x_2 = 12, \\ 2x_1 + x_2 = 1. \end{cases}$$

正常使用主观题需2.0以上版本雨课堂

作答



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例1 求解二元线性方程组
$$\begin{cases} 3x_1 - 2x_2 = 12, \\ 2x_1 + x_2 = 1. \end{cases}$$

解 由于
$$D = \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = 3 - (-4) = 7 \neq 0,$$

$$D_1 = \begin{vmatrix} 12 & -2 \\ 1 & 1 \end{vmatrix} = 12 - (-2) = 14,$$

$$D_2 = \begin{vmatrix} 3 & 12 \\ 2 & 1 \end{vmatrix} = 3 - 24 = -21,$$

因此
$$x_1 = \frac{D_1}{D} = \frac{14}{7} = 2, \quad x_2 = \frac{D_2}{D} = \frac{-21}{7} = -3.$$

二、三阶行列式

三元线性方程组
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \neq 0$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3; \end{cases}$$

若记

$$D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix} = b_1 a_{22} a_{33} + b_2 a_{13} a_{32} + b_3 a_{12} a_{23} \\ - b_1 a_{23} a_{32} - b_2 a_{12} a_{33} - b_3 a_{22} a_{31}$$

或

$$\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \quad D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix},$$

$$D_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \quad D_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}.$$

则三元线性方程组的解为：

$$x_1 = \frac{D_1}{D}, \quad x_2 = \frac{D_2}{D}, \quad x_3 = \frac{D_3}{D}.$$

例1 解线性方程组

$$\begin{cases} x_1 - 2x_2 + x_3 = -2, \\ 2x_1 + x_2 - 3x_3 = 1, \\ -x_1 + x_2 - x_3 = 0. \end{cases}$$

解 由于方程组的系数行列式

$$D = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -3 \\ -1 & 1 & -1 \end{vmatrix} = 1 \times 1 \times (-1) + (-2) \times (-3) \times (-1) \\ + 1 \times 2 \times 1 - 1 \times 1 \times (-1) - (-2) \times 2 \times (-1) - 1 \times (-3) \times 1 \\ = -5 \neq 0,$$

同理可得

$$D_1 = \begin{vmatrix} -2 & -2 & 1 \\ 1 & 1 & -3 \\ 0 & 1 & -1 \end{vmatrix} = -5, \quad D_2 = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -3 \\ -1 & 0 & -1 \end{vmatrix} = -10,$$

$$D_3 = \begin{vmatrix} 1 & -2 & -2 \\ 2 & 1 & 1 \\ -1 & 1 & 0 \end{vmatrix} = -5,$$

故方程组的解为:

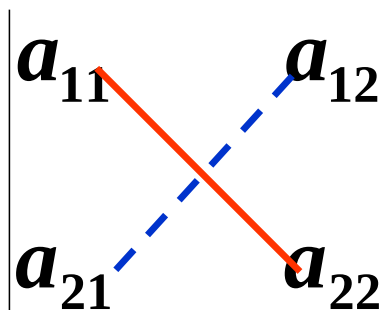
$$x_1 = \frac{D_1}{D} = 1, \quad x_2 = \frac{D_2}{D} = 2, \quad x_3 = \frac{D_3}{D} = 1.$$

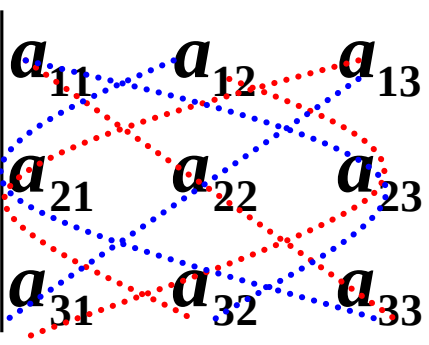


问题：什么是 n 阶行列式？

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = ?$$

回忆


$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$


$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$


$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = ?$$

特点: 1. n^2 个元素

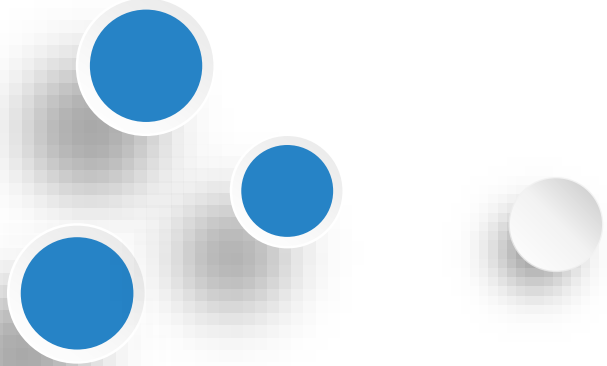
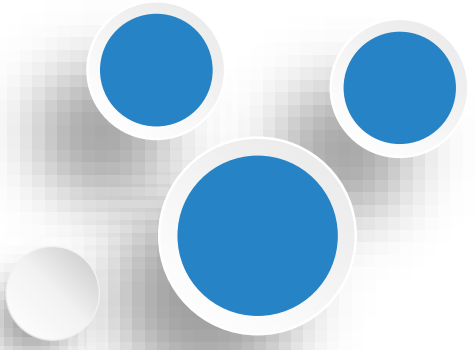
2. 共有 $n!$ 项代数和

3. 每项为取自不同行不同列的元素乘积

4. 正负项的个数相等

5. ?, 取正号

?, 取负号



§ 2

排列

§2 全排列与逆序数



1.排列的相关概念

2.排列的奇偶性

1.排列的相关概念



n 级排列: 由 $1, 2, \dots, n$ 组成的一个有序数组.

自然排列 (标准次序): $123\dots n$

逆序: 在一个排列中, 如果一对数的前后位置与大小顺序相反

逆序数: 排列中逆序的总数.



(1)

3	2	5	1	4
↓	↓	↓	↓	↓
0	1	0	3	1

于是排列32514的逆序数为

$$t = 0 + 1 + 0 + 3 + 1 = 5.$$


$$(2) \quad n(n-1)(n-2)\cdots 321$$

解

$$n(n-1)(n-2)\cdots 321$$

$$\begin{aligned} t &= 1 + 2 + \dots + (n-2) + (n-1) \\ &= \frac{n(n-1)}{2}, \end{aligned}$$

2.排列的奇偶性



奇排列：逆序数为奇数的排列

偶排列：逆序数为偶数的排列

对换： $p_1 \cdots p_i \cdots p_j \cdots p_n \longrightarrow p_1 \cdots p_j \cdots p_i \cdots p_n$

相邻对换： $p_1 \cdots p_i p_{i+1} \cdots p_n \longrightarrow p_1 \cdots p_{i+1} p_i \cdots p_n$



定理1：对排列进行一次对换则改变排列的奇偶性.



定理1: 对排列进行一次对换则改变排列的奇偶性

推论1: 奇（偶）排列调为自然排列所需的对换次数
为奇（偶）数

推论2: 所有 n 元排列（ $n>1$ ）中奇偶排列各半